

Qingyang Zhou

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<https://esperarr.github.io/CV/>

EDUCATION

Tsinghua University September 2022 - Present
Xinya College, Tsinghua University: <https://www.xyc.tsinghua.edu.cn/index.htm>
Department of Computer Science and Technology, Tsinghua University: <https://www.cs.tsinghua.edu.cn/>
GPA 3.6/4.0
Selected A+ professional compulsory course: Introduction to Complex Analysis, Media Computing, Database Special Topic Training;

RESEARCH PUBLICATIONS

"Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation"

Yuwei Zhou, Xin Wang, Hong Chen, Yipeng Zhang, **Qingyang Zhou**, Wenwu Zhu
Submitted for publication

"Towards Mutually Illuminative Collaborative Human-AI Deliberate Discussion"

Zipeng Zhang, Shiwei Wu, Ran Ran, **Qingyang Zhou**, Ranrui Ma, Jiaye Leng, Jian Zeng, Zhenhui, Chun Yu
Submitted for publication

INTERNSHIP/TRAININGS

Client Development Engineer Intern at Tencent (Project MAX) May 2026 - August 2026
Tencent, Project MAX Team

- Optimized NavMesh pathfinding to improve agent movement smoothness, resolving instability on terrain bumps and convex corner edges; implemented NPC auto-navigation using custom FixNavMesh algorithms.
- Developed a GM panel for debugging and performance stats toggling, and built the game control panel UI for the 2V2 battle mode.

scientific research internship at Multimedia and Network Big Data Lab July 2023 - July 2024

Tsinghua University Multimedia and Network Big Data Lab: <https://mn.cs.tsinghua.edu.cn/index/index.html>

- Conducted study and research on diffusion models and personalized image generation, within the project "Video Generation System Combining Large Language Models with Diffusion Models".
- Assisted in publishing the paper: "Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation".

scientific research internship at Pervasive Human Computer Interaction Lab January 2025 - April 2025

Tsinghua University Pervasive Human Computer Interaction Laboratory: <https://pi.cs.tsinghua.edu.cn/>

- Conducted research on aligning human and AI thought processes.
- Developed a human-AI interactive writing assistant to support collaborative idea generation.
- Contributed to the research and UIST 2025 submission of the paper "Towards Mutually Illuminative Collaborative Human-AI Deliberate Discussion".

scientific research internship at Information Retrieval Lab October 2024 - Present

Information Retrieval Lab at Tsinghua University: <http://www.thuir.cn/>

- Focused on improving traditional recommendation models (such as FM, DeepFM, AFM).
- Incorporated SAE plugins into these models to enhance their interpretability, controllability, and transparency.

POSITION OF RESPONSIBILITY

Secretary of the Intelligent Agent Department of the Student Science and Technology Association

February 2023 - Present

Tsinghua University CST Department, Student Science and Technology Association: <https://net9.org/home/>

- Participated in the game development and event organization for the Intelligent Agent Competition.
- Mainly responsible for the development of the Intelligent Agent-Based Game 'Generals' Project in the 28th Tsinghua University Intelligent Agent Competition.

Volunteer Tutor at Tsinghua Q&A Workshop

September 2025 - Present

Tsinghua University, Center for Student Learning and Development: <https://learning.tsinghua.edu.cn/xgk/ywjs.htm>

- Provided tutoring for programming and computer science courses, including C/C++ programming, data structures, and algorithms.
- Assisted students with debugging, algorithm design, and understanding core programming concepts during regular tutoring sessions.

Tutor of Tsinghua University Science and Engineering Summer Camp

July 2024 - August 2024

Tsinghua University 2024 Science and Engineering Summer Camp I6 Class Tutor

- Responsible for event organization, answering professional questions, and other related tasks.
- Led and organized engineering-related activity projects, such as intelligent robot design and card-bridge construction.

Educational Outreach & Community Service

2022 - Present

Tsinghua University Public Welfare Initiatives

- Exchanged letters with and donated books to children in rural mountainous areas, offering ongoing encouragement and mentorship.
- Delivered lectures at a county high school on study methods and university major selection during rural field practice; advised prospective students on academic pathways as part of Tsinghua's Tianjin recruitment team.

PROJECTS

Path Tracing Renderer

June 2026

Personal Graphics Project: <https://github.com/Esperarr/TinyPathTracer>

- A physically-based path tracer supporting Whitted-style ray tracing and Monte Carlo path tracing, with PBR materials (GGX importance sampling, Cook-Torrance BRDF, Fresnel Schlick) across diffuse, mirror, glass, and glossy surfaces.
- Features BVH acceleration for 100k+ face meshes, NEE + MIS, depth of field, MSAA anti-aliasing, gamma correction, and OpenMP parallelization.

Intelligent Agent-Based Game 'Generals'

September 2023 - May 2024

Major Project in the 28th Tsinghua University Intelligent Agent Competition: <https://www.saiblo.net/game/35>

- A turn-based strategy game developed for the competition; responsible for backend logic, unit tests, debugging, SDK creation, documentation, and guiding participants.

SKILLS, INTERESTS & EXTRA-CURRICULAR

Languages

Chinese (Native), English (Fluent), Spanish (Basic)

Programming

C/C++, Python, JavaScript/TypeScript, Java, Rust, Verilog/SystemVerilog

Graphics / Tools

OpenGL, GLSL, Unity, PyTorch, Git, Linux, Docker, LaTeX

Writing & Arts: Passionate about creative writing (essays and poetry); member of the university Musical Theatre Club, Classical Poetry Society, and Street Dance Club.

Student Service: Publicity Department Executive at Xinya College (Sept 2022 - Sept 2023).